

WHAT LIMITS THIS QUBIT'S COHERENCE AND WHAT I WOULD FIX FIRST

Experimental Results:

Experiment	Simulation parameters	Estimated values
T1 Experiment	$T1=50\mu\text{s}$, $T2=60\mu\text{s}$	$T1 = 49.580 \pm 0.553\mu\text{s}$
Ramsey experiment	$T1=80\mu\text{s}$, $T2=30\mu\text{s}$	$T2^* = 28.684 \pm 0.817\mu\text{s}$
Hahn echo	$T1=100\mu\text{s}$, $T2^*=50\mu\text{s}$, $T\emptyset=150\mu\text{s}$	$T2^*=59.071 \pm 0.364\mu\text{s}$ $T2_echo=99.434 \pm 0.797\mu\text{s}$ $T2_echo/T2^* = 1.68$

Analysis of Experimental Results:

T1 Experiment:

The close agreement between the input T1 and the fitted T1 indicates that the simulated inversion-recovery experiment and the exponential fitting procedure successfully recovered the relaxation time within the fitting uncertainty.

Ramsey Experiment:

Ramsey experiment and damped cosine fitting procedure successfully recovered the qubit's coherence time within the fitting uncertainty

Observation : here $T2^* < T1$ obviously as $T2^*$ is combination of both energy relaxation and dephasing where T1 is only energy relaxation.

Hahn echo experiment:

$T2_{echo} > T2^*$ indicates that Hahn Echo model built successfully refocuses slow frequency fluctuations (quasi-static dephasing noise), reducing phase errors and extending the qubit coherence time.

$T2_echo/T2^* = 1.68$ (ratio > 1) → confirms that some of the dephasing in this simulated qubit is quasi-static in nature. And (1.68, not much larger) moderate size of the ratio suggests quasi-static and dynamic dephasing contribute comparably, rather than one totally dominating.

DOMINANT NOISE CHANNEL:

With $T2_echo/T2^* = 1.68$, the echo pulse recovered noticeable but not dramatic coherence — meaning quasi-static noise (like flux noise) contributes meaningfully, but is not the single overwhelming factor. A large chunk of dephasing still comes from the dynamic (Markovian) part, which the echo cannot cancel.

So the dominant noise channel here looks like a mix of flux noise (quasi-static) and fast dephasing sources like TLS defects (dynamic), with neither one overwhelmingly dominant — unlike a very high ratio (e.g. 3-4x+), which would point almost entirely to quasi-static/flux noise alone.

WHAT I WOULD FIX FIRST:

Since flux noise appears to be the larger contributor, I would first operate the qubit at its flux "sweet spot" — the bias point where the qubit frequency is least sensitive to flux fluctuations ($\partial\omega_q/\partial\Phi = 0$). This directly suppresses the dominant quasi-static dephasing channel with no hardware redesign needed. For the remaining TLS-driven dynamic dephasing, longer-term improvements would target junction and substrate interface quality during fabrication to reduce defect density.

CONCLUSION:

Across all three experiments, the simulation pipeline correctly recovered the assumed ground-truth T_1 and T_2^* values, validating the fitting procedures. The key diagnostic result came from comparing T_2^* and T_{2_echo} within the Hahn Echo experiment: a ratio of 1.68 indicates the qubit's coherence is limited by a mix of quasi-static dephasing (likely flux noise) and fast dynamic dephasing (TLS defects), with flux noise being the somewhat larger contributor, while T_1 relaxation is not the bottleneck. The most practical first fix is operating at the flux sweet spot, with fabrication improvements to reduce TLS defects as a longer-term step.